



COMMENTARY

Ancestral gene regulatory networks drive cancer

Kimberly J. Bussey^{a,b}, Luis H. Cisneros^{a,c}, Charles H. Lineweaver^d, and Paul C. W. Davies^{c,1}

Although cancer is one of the most intensively studied phenomena in biology and occurs in almost all multicellular species (1, 2), an explanation for its existence and properties within the context of evolutionary history has received comparatively little attention. However, it is widely recognized that progress in treatment and prevention depends on a deeper understanding of the biology of cancer (3).

Many of the hallmarks of cancer (4, 5) are reminiscent of unicellular life, suggesting that neoplasms represent a type of throwback or reexpression of ancestral traits. Theodore Boveri first suggested that cancer recapitulates ancient phenotypes (6). This basic idea has

recently been developed into the atavistic theory of cancer, which seeks to trace cancer's deep evolutionary roots to make specific predictions about gene expression in tumorigenesis (7, 8). The atavistic theory postulates that the biological origin of cancer can be found in the early transitional phase from unicellularity (UC) to multicellularity (MC), before the emergence of complex metazoans about 600 Mya. These ancestral traits reappear because the regulation that suppresses them or restricts them to specific contexts (e.g., embryogenesis or wound-healing) becomes disrupted. In broad terms, the atavistic theory predicts up-regulation of genes with UC evolutionary origins and down-regulation of genes that evolved after the advent of MC (Fig. 1). The work of Trigos et al. (9), reported in PNAS, sets out to test this prediction.

Trigos et al. (9) studied gene expression in seven types of solid tumors and investigated the corresponding gene ages using a method known as phylostratigraphy (10, 11). Specifically, the authors (9) assigned genes to one of 16 phylostrata and then compared gene expression in tumors with that in normal tissue. They found that tumors overexpress more genes from the first two (oldest) phylostrata compared with normal tissues, whereas genes from evolutionarily more recent phylostrata 3–12 are generally underexpressed in tumors. These results support the atavistic theory of cancer. Another prediction of the theory is that as cancer progresses from low to high grade, the loss of differentiation can be understood as older genes being expressed at higher levels. In figure 1D of Trigos et al. (9), the authors present evidence supporting this prediction, showing an inverse relationship between Gleason score and transcriptome age index (a measure of gene expression weighted by age) in prostate cancers. Using their ages to classify UC versus MC process genes, Trigos et al. observed that in general those cellular processes assigned to a UC origin were more active in tumors. Although increased expression of UC genes seems to be a general phenomenon, specific up-regulation depends on the function of the associated processes, suggesting

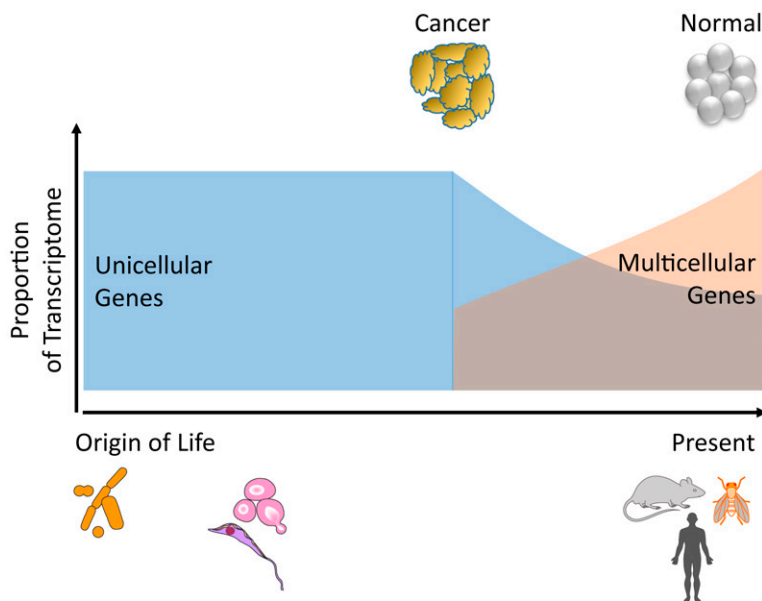


Fig. 1. Compared to normal, cancer increases the proportion of its transcriptome coming from unicellular genes. The atavistic theory postulates that cancer results from the inappropriate reexpression of an ancient toolkit of adaptations suited to UC colonial life and subsequently suppressed in the context of complex MC life. One prediction from this theory is that cancer will overexpress genes and processes with a UC origin, while simultaneously underexpressing the MC pathways that promote cellular cooperation for the good of the organism. This prediction is confirmed in the analysis of Trigos et al. (9).

^aNantomics, LLC, Tempe, AZ 85281; ^bDepartment of Biomedical Informatics, Arizona State University, Tempe, AZ 85287; ^cBEYOND Center for Fundamental Concepts in Science, Arizona State University, Tempe, AZ 85287; and ^dPlanetary Science Institute, Research School of Astronomy and Astrophysics and Research School of Earth Sciences, Australian National University, Canberra, ACT 2611, Australia

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¹To whom correspondence should be addressed. Email: Paul.Davies@asu.edu.

that cancer atavism is not a haphazard affair but an ancient choreography of correlated gene expression.

Supporting evidence for an organized pattern of expression comes from an analysis of coregulation based on evolutionary age. First, the correlation of expression between pairs of UC processes or MC processes were generally positive in both tumor and normal samples, suggesting a compartmentalization of processes with different evolutionary histories as a general feature across physiology. However, cross-correlations between UC and MC processes were significantly more negative in tumors than in the normal samples. Second, the amount of variability within pairwise correlations was reduced in tumors. Finally, in a comparison of normal and tumor samples, 20 of 24 UC–MC correlations switched from positive to negative, whereas only 4 switched in the opposite direction. Almost half of the correlations (11 of 24) were associated with cell death. Trigos et al. (9) hypothesize that the disruption of cross-talk between older (UC) and younger (MC) processes may prove to be a key factor in the onset of tumorigenesis.

To investigate that idea, Trigos et al. (9) identified 12 genes that represent consistent hubs across the seven tumor types studied. The authors note that four genes were associated with Rap 1 signaling. Rap 1 is a small GTPase involved in coordinating cell cycle and cell-cell adhesion with environmental signals. A quick annotation of the 12 genes using DAVID (<https://david.ncifcrf.gov/gene2gene.jsp>) demonstrates that 11 are involved in cytoskeletal signaling and actin dynamics. Applying DAVID to the total list of 103 genes (12 discussed in the main paper of ref. 9 and 91 in the Supporting Information), reveals significant enrichment in ribosomal biogenesis, urea cycle regulation, RNA catabolic pathways, and glycolysis. Also highly enriched are the so-called 14-3-3 proteins, which lie downstream of Rap1 and serve to modulate the effect of many receptor tyrosine kinase signaling cascades, including those that activate MEK and PI3K. These are two major pathways implicated in tumorigenesis and the focus of many targeted therapeutics used clinically or currently in clinical development. All of these functions have been implicated in cancer biology.

The discovery that cancer reduces the links between gene regulatory networks defined by evolutionary history has important clinical consequences. Recent work by Wu et al. (12) investigated the acquisition of drug resistance in response to drug gradients, and demonstrated that resistance evolved from the up-regulation—and not mutation—of UC genes. The key to applying this knowledge for therapeutic benefit lies in determining which processes or interactions, UC or MC, are likely to provide the largest therapeutic window. In the late 1980s and early 1990s the field of tumor-suppressor studies yielded many examples where introduction of an intact copy of the tumor suppressor *de jour* resulted in diminished tumorigenic potential (reviewed in ref. 13). Using such methods, it is fairly easy to suppress rapid cell division, migration, and invasion in cancer cells in vitro and in xenografts. However, similar success in a clinical setting has proved elusive. It is much easier to turn off a pathway that has been inappropriately activated than to restore a link that has been severed by damage. This is where appeal to

the atavistic theory could prove therapeutically significant. Analyses like that of Trigos et al. (9) offer the opportunity to design synthetic lethality screens that take advantage of cancer's atavistic switch to more primitive ancestral phenotypes, by identifying those links at the UC–MC interface where the tumor microenvironment places differential demands and selective pressures on tumor cells compared with normal cells (8). An example is the use of hypoxia to induce "BRCA-ness," which causes sensitivity to poly(ADP-ribose) polymerase inhibition, in combination with DNA-damaging agents that promote double-strand breaks in tumors, independently of their BRCA1/BRCA2 mutational profile (14). This strategy has been

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successful in preclinical experiments because hypoxia down-regulates the expression of homologous recombination repair. Because of their reliance on nonhomologous recombination-mediated pathways of double-strand break repair, tumors are thus sensitized to agents that inhibit those pathways, whereas normal tissue is unaffected.

Another application of the atavism theory suggests that genomic instability, one of the best-known hallmarks of cancer, derives from an atavistic reexpression of a stress-induced mutational response of the sort observed in bacteria (15). The analysis presented by Trigos et al. (9) offers the opportunity to identify the links between the UC processes involved in this response and the MC processes that regulate it, allowing the design of therapeutic regimens that take advantage of the difference in the rates of evolution between tumors and the surrounding microenvironment.

Trigos et al. (9) present the most comprehensive evidence that a general shift to the preferential expression of more ancient genes is a common feature of tumors, providing substantial support for the atavistic theory. To test the theory further, we suggest a proteomics-based study across cancer types and normal tissues with an analysis using more phylostrata to determine how the observed alterations reported by Trigos et al. translate into changes in physiology, as proteins are generally the direct targets of therapy and gene expression does not always correlate with protein expression (16). Another outstanding question is whether the atavistic switch is reversible or more akin to a permanent speciation event. The challenge is to convert insights provided by the atavism theory into improved clinical outcomes (8). Such an understanding has implications for interpreting the evolution of cancer in response to therapy, countering phenomena such as drug resistance and immune evasion, as well as developing strategies to improve prevention.

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